

Indian Wedding Cost Analysis

Data Analysis Using Python

**Prepared By:
Yashodhan Kapse**

Tools Used:

Python

Pandas

NumPy

Matplotlib

Seaborn

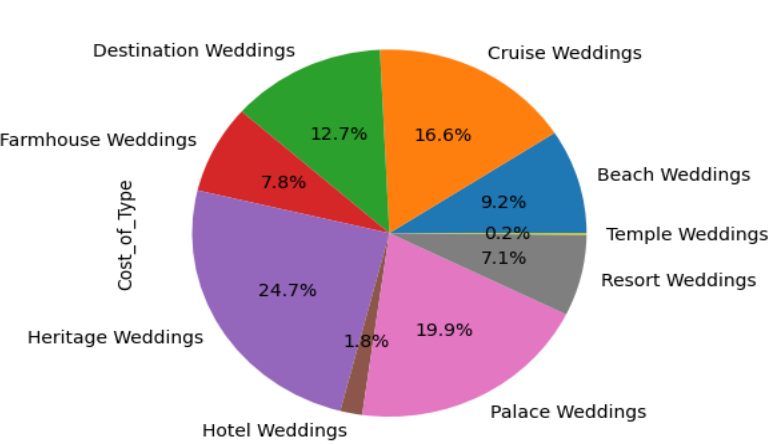
Jupyter Notebook

Dataset: Indian Wedding Cost Dataset

Project Objective:

To analyze wedding costs, decoration expenses, entertainment costs, and wedding types using data visualization techniques.

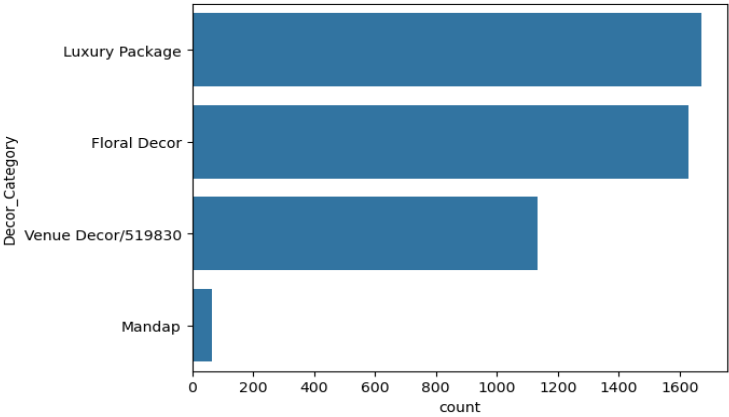
Cost Distribution by Wedding Type (Pie Chart)



Observations:

- Heritage Weddings have the highest share of the total wedding cost at 24.7%.
- Palace Weddings are the second highest contributor with 19.9%.
- Cruise Weddings account for 16.6% of the total cost.
- Destination Weddings contribute 12.7%, while Beach Weddings contribute 9.2%.
- Farmhouse Weddings (7.8%) and Resort Weddings (7.1%) have moderate cost shares.
- Hotel Weddings contribute only 1.8% of the total cost.
- Temple Weddings have the lowest contribution at just 0.2%, making them the least expensive wedding type.
- Overall, luxury wedding types such as Heritage, Palace, and Cruise Weddings dominate the total wedding expenditure.

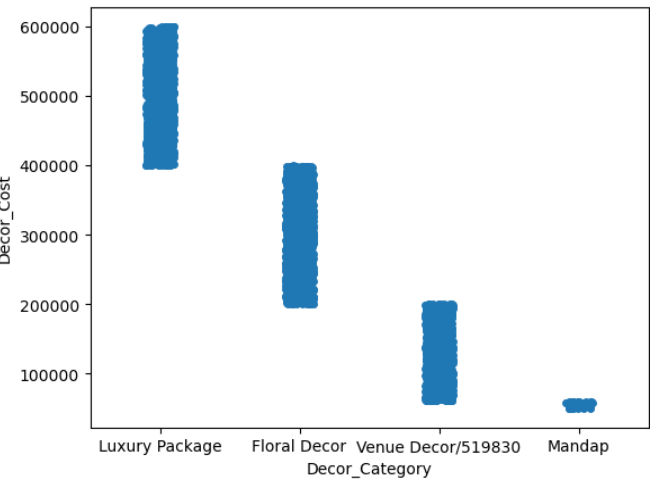
Count Plot of Decor_Category



Observations:

- Luxury Package has the highest number of records (around 1700), making it the most common décor category.
- *Floral Decor is the second most frequent category (around 1650 records). Venue Decor has a moderate number of records (around 1150).
- Mandap has the lowest frequency (less than 100 records), indicating it is the least represented category.
- *The distribution of décor categories is uneven, with Luxury Package and Floral Decor dominating the dataset.

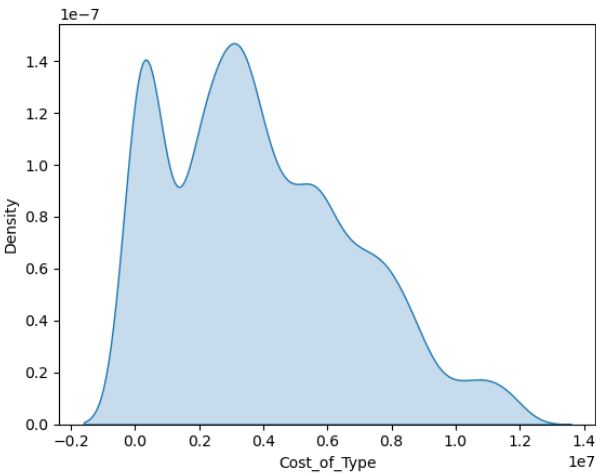
Decor Cost by Decor Category (Strip Plot)



Observations:

- Luxury Package has the highest decor cost, ranging from approximately ₹4,00,000 to ₹6,00,000.
- *Floral Decor has a medium cost range of about ₹2,00,000 to ₹4,00,000.
- Venue Decor has a lower cost, mostly between ₹60,000 and ₹2,00,000.
- Mandap has the lowest decor cost, around ₹50,000 to ₹70,000.
- The plot shows a clear decrease in decor cost from Luxury Package to Mandap.
- Luxury decor options are significantly more expensive than basic decor categories.

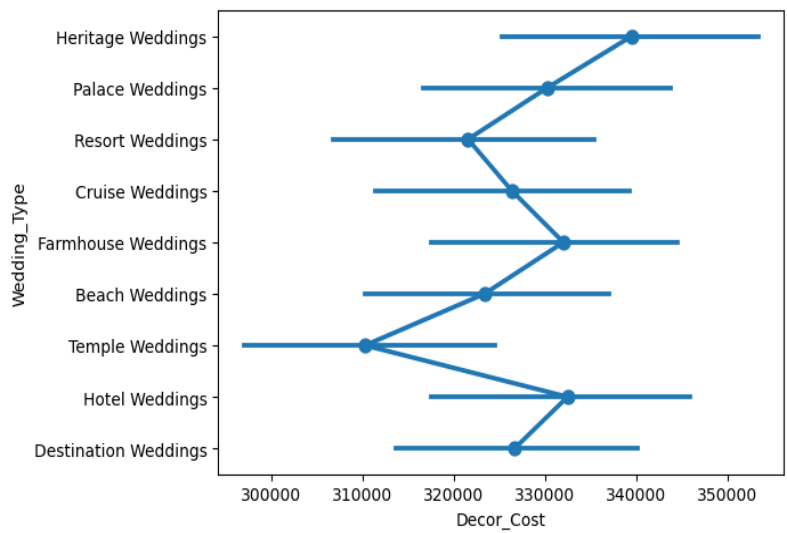
KDE Plot of Cost_of_Type



Observations:

- The Cost_of_Type distribution is positively (right) skewed, with most values concentrated in the lower to middle cost range.
- The plot shows multiple peaks (multimodal distribution), indicating different groups or categories of wedding costs.
- The highest density is observed around the ₹2–4 million (20–40 lakh) range, where most weddings fall.
- The KDE plot reveals that wedding costs are concentrated in the mid-range, with a right-skewed and multimodal distribution. Most weddings have moderate costs, while only a small number belong to the high-cost luxury segment.

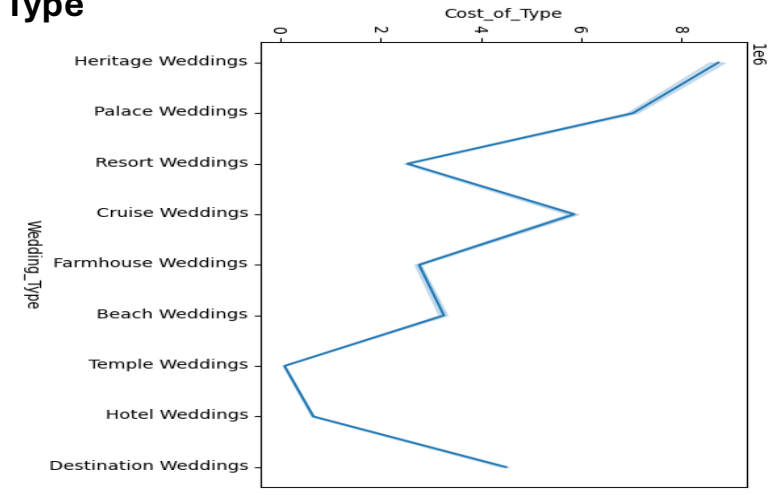
Point Plot of Decor Cost by Wedding Type



Observations:

- The analysis shows that decor costs vary slightly across wedding types, with Heritage Weddings incurring the highest average decor expenses and Temple Weddings the lowest. Most wedding categories have decor costs clustered between ₹320K and ₹335K, indicating relatively stable decoration spending across different venue types.
- The Heritage Weddings category has the highest average decor cost (around ₹340K), indicating premium decoration requirements.
- Temple Weddings have the lowest average decor cost (around ₹310K), making them the most budget-friendly option.

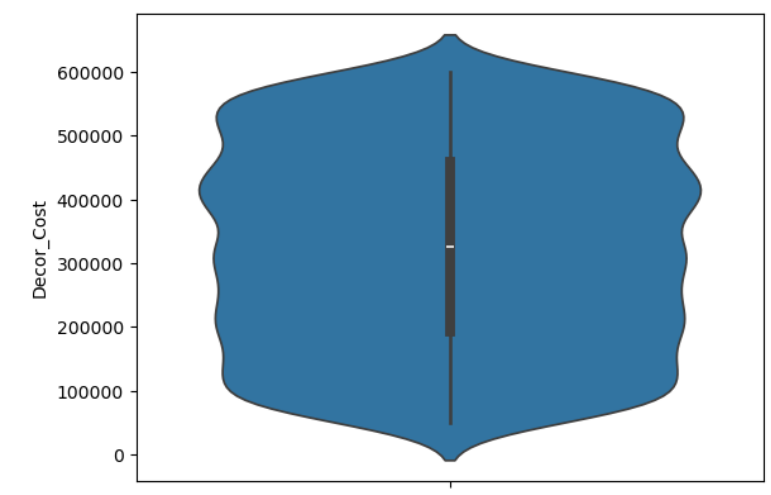
Line Plot – Average Cost of Type by Wedding Type



Observations:

- Heritage Weddings have the highest average Cost_of_Type.
- Temple Weddings have the lowest average Cost_of_Type.
- Palace Weddings also show relatively high costs.
- Cruise Weddings are costlier than Resort, Beach, and Farmhouse Weddings.
- Most wedding types have moderate average costs, with only a few premium categories showing significantly higher values.
- Overall, premium wedding types require higher budgets than traditional or budget-friendly wedding types.

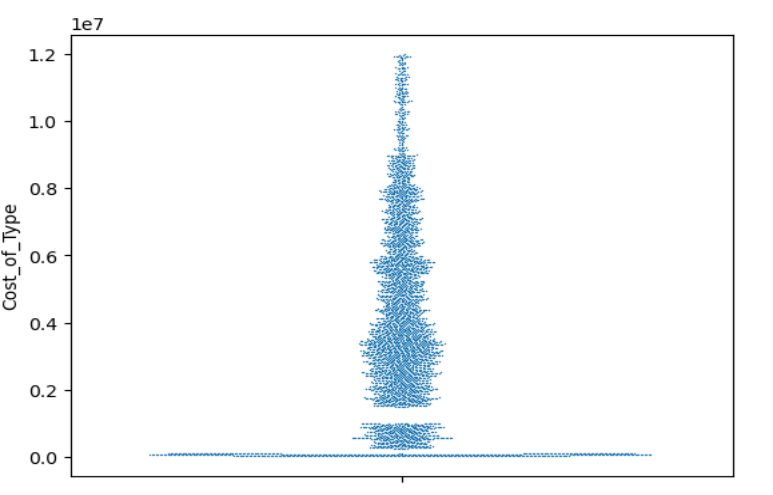
Violin Plot of Decor Cost



Observations:

- The violin plot shows that decor costs are widely distributed, with the majority of weddings spending between ₹200K and ₹500K.
- The median is around ₹325K, and the absence of significant outliers suggests a balanced distribution of decoration expenses across the dataset.
- The highest density of observations lies between ₹200,000 and ₹500,000, meaning most weddings fall within this cost range.
- The distribution appears fairly balanced, with no strong skew toward either low or high decor costs.

Line Plot – Average Cost of Type by Wedding Type



Observations:

- Heritage Weddings have the highest average Cost_of_Type.
- Temple Weddings have the lowest average Cost_of_Type.
- Palace Weddings also show relatively high costs.
- Cruise Weddings are costlier than Resort, Beach, and Farmhouse Weddings.
- Most wedding types have moderate average costs, with only a few premium categories showing significantly higher values.

Pivot Table – Average Decor Cost by Wedding Type and Decor Category

Decor_Category	Floral Decor	Luxury Package	Mandap	Venue Decor/519830	All
Wedding_Type					
Beach Weddings	298666.468421	500116.769663	56033.000000	123695.531746	323378.498000
Cruise Weddings	304377.240642	498850.365169	56994.000000	131126.196970	326387.098000
Destination Weddings	297909.338624	494772.576087	55553.700000	131988.555556	326682.434000
Farmhouse Weddings	306232.218391		54932.142857	129500.253968	332039.752000
Heritage Weddings	299675.789773	501587.887255	54077.666667	126049.596491	339521.976000
Hotel Weddings	303936.565217	498630.500000	55184.833333	127091.900000	332492.520000
Palace Weddings	296340.389535	496876.517766	53482.600000	127104.928571	330275.710000
Resort Weddings	295525.321429	499619.886486	56616.750000	131353.345324	321577.964000
Temple Weddings	295854.767196	495173.672840	55355.687500	136265.827068	310287.464000
All	299850.867403	498195.349491	55356.044776	129453.981465	326960.379556

Observations:

- Luxury Package has the highest average cost across all wedding types (around ₹4.95–5.01 lakh), making it the most expensive decor category.
- Mandap is the least expensive decor category, with average costs around ₹53K–57K.
- Venue Decor costs remain relatively consistent across wedding types, averaging around ₹1.23–1.36 lakh.
- Floral Decor averages approximately ₹2.96–3.06 lakh, showing only minor variation between wedding types.
- Based on the overall average decor cost (All), Heritage Weddings have the highest average decor expense (~₹339,522), while Temple Weddings have the lowest (~₹310,287).
- The small differences across wedding types indicate that decor category has a greater impact on total decor cost than the wedding type itself.

Insight

The pivot table shows that Luxury Packages consistently contribute the highest decoration costs across all wedding types, while Mandap decoration remains the least expensive. Heritage Weddings incur the highest overall average decor cost, whereas Temple Weddings are the most economical. Overall, decor category influences the total decoration budget more significantly than the type of wedding.

Conclusion

Key Findings

Heritage, Palace, and Cruise Weddings have the highest overall costs.

Luxury Package is the most expensive decor category.

Mandap decoration has the lowest cost.

Wedding expenses vary significantly across different categories.

Data visualization helps identify cost trends and spending patterns effectively.

Future Scope

Build an interactive Power BI dashboard.

Predict wedding costs using Machine Learning.

Compare costs across different cities and states.

Develop a wedding budget planner application.

Thank You

Prepared By:

Yashodhan Kapse